B V N S HEMANTH

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AI engineer building **production LLM agents** — currently first employee at an AI-analytics startup where my LLM data-analyst agent is **live with paying clients**. B.Tech from IIT Madras (**Minor in AI/ML**) with a **speech-technology** foundation; filed **2 UK patents**, placed in the top **5%** of the Shell AI Hackathon. Deployed real-world systems across **LLM agents, evaluation/validation, reinforcement learning, PINNs, and speech systems**.

EDUCATION

Program	Institution	%/CGPA	Year of Completion
B.Tech (Mechanical Engineering)	Indian Institute of Technology, Madras	8.25/10	2024
XII (Andhra Pradesh State Board)	Aditya Junior College, Punadipadu	97.4%	2020
X (Andhra Pradesh State Board)	Narayana E.M. High School, Kannur	10/10	2018

SCHOLASTIC ACHIEVEMENTS

- **AIR 4457** among **150k+** students in JEE Advanced *(2020)* | **AIR 1855** among **1 million** students in JEE Mains in first attempt *(2020)*
- **One** among **1145 out of 50k+** students promoted to the **second level**, Hero Campus Case Study Challenge Season 7 *(2021)*
- Stood **second** out of **500+** participants in Mechanical Trivia **Quiz** *(2021)* | Ranked **347** in AP EAMCET out of **200k+** students *(2020)*

RELEVANT COURSEWORK AND SKILLS

Pattern Recognition and Machine Learning			Speech Technology		ML for Engineering and Science	
Probability, Statistics & Stochastic Processes			Multi-armed Bandits		Data Structures & Algorithms	
Languages & Software	Python	C, C++	SQL	Matlab	VBA	Git
Skills	LLM Agents	Speech Tech	Evaluation / Validators	RL	PINNs / Neural Operators	GANs
Frameworks	PyTorch	HuggingFace	FastAPI	AWS	Scikit-learn	Pandas / Numpy
Math	Probability	Statistics	Optimization	Linear Algebra	Numerical Methods	Calculus

PROFESSIONAL EXPERIENCE AND RESEARCH WORK

infigro.ai
Data Scientist & AI Agent Developer
First Employee
(Feb '26 – Now)

Trustworthy AI Analytics — LLM Data-Analyst Agent

- First engineering hire; own the product's differentiating IP layer end-to-end: analytical **methods**, runtime **validators**, and **evaluation** for an LLM-powered data-analysis agent.
- Designed the **multi-stage agent orchestration** (planner → analysis engines → validators → renderer): EDA inference engine, decomposition/composition-effect guards, and **causal-honesty rules** that constrain what LLM-generated analysis may claim from data.
- Re-architected the data layer from CSV/pandas to **Parquet**, cutting end-to-end pipeline runtime from **40 → 15 minutes (2.7×)**.
- Designed the **prompt architecture** for the agent: typed output contracts, contrast-based instruction over cloneable examples, and per-question-class voice control.
- Built **deterministic repair and graceful-degradation paths** so malformed LLM outputs self-correct at runtime instead of failing client runs.
- Built **deterministic grounding validators** enforcing analytical correctness on every generated output at runtime — numbers, filters, and claims verified against source data before delivery.
- Proved validator reliability via **mutation testing**, golden regression suites, and inter-stage handshake/render verifiers — seeded known defects into outputs to verify the checks themselves catch them.
- Own the **full data pipeline end-to-end**: ingestion and sanitization, EDA, statistical insight generation, chart/report rendering, and delivery — plus operator tooling for rapid iteration on live outputs.
- Designed the **statistical methodology** behind automated insights: decomposition analysis, mix-shift/composition-effect handling, and robustness checks on every claim.
- Shipped **release v1 to production (Sep '26), live with 2 clients and 2 more signed on annual agreements**; built the multi-grain chart/report layer (ECharts/Plotly) with question-class-aware narrative voicing across **70+ analysis lanes**.
- Operate beyond engineering as first employee: **mentor a junior engineer** on pipeline integration, participate in **client and business meetings**, translate business questions into

	analytical requirements, and helped drive **product pivots** from live client feedback.
Jaguar Land Rover **Graduate Engineer** *(Jul '24 – Apr '26)*	***R&D Engineer – Recruiting Tool with LLMs*** • Built **solo LLM pipeline** (Flan-T5-XXL) to bucket/deduplicate 3,000 questions with **100% accuracy**. • Eliminated **~9M** manual comparisons, saving **100+ hours/run** and **₹5 lakh+** costs; **piloted** in recruitment workflows, deployed weekly during hiring cycles. ***R&D Engineer – PINNs & Neural Operators*** • Replaced FEM/CFD sims using PINNs, achieving **18× faster inference (~5s vs 90s)** with error **≈10^{-5}**; implemented loss balancing, Fourier embeddings, adaptive collocation in **PyTorch**. • Drafted paper on **PINNs for FGM** (conference submission in progress); presented internally, secured **4th/40**; awarded AWS credits + conference sponsorship. ***R&D Engineer – Requirements & Test Case Automation*** • Solo project using **OCR/diagram parsing** to auto-generate requirements/test cases in EARS syntax; reduced manual effort from **days** → **minutes**; tool under experimental adoption across teams. ***Mechanical Software Development Engineer – Longitudinal SW Function (Roll Stability Control)*** • Modernized **100+ legacy C** files introducing 2D/3D datatypes; built Python automation tool parsing DCM files, eliminating VBA dependency and saving **70%** of time; work integrated into RSC pipeline, scheduled for **2027** cars. ***White Paper – Generative AI in Business Applications*** • Authored and presented a white paper on GenAI business use cases; recognized by **senior stakeholders** for actionable insights and strategy alignment.
Freelance Data Scientist & AI Engineer — Upwork *(2025 – Present)*	• Automated **2D → 3D building & floor plan conversion** with interior layouts and regulatory compliance; deployed via REST APIs for client use. • Built and deployed **LLM-powered automation workflows** (text clustering, deduplication, semantic search) saving clients **100+ hours/run**. • Developed custom data analysis dashboards (**Python, FastAPI, AWS**) enabling non-technical stakeholders to extract insights seamlessly.
PATENTS & PUBLICATIONS	
Patents	• **Patent (Filed, UK 2024)** – Simulated Driving Experience using Integrated Vehicle Systems (Co-Inventor) • **Patent (In Filing, 2025)** – Steer-for-Traction Algorithm (Co-Inventor)
Publications	• **Publication (In Preparation)** – PINNs for Functionally Graded Materials • **Journal Publication (In Preparation)** – A robust formulation of a PINN for flow over airfoil
HACKATHONS	
Shell (AI Hackathon) *Fuel Blend Property Prediction* *(Top 5%, Score: 86.64/92)*	• Engineered multi-stage ML pipelines (**CatBoost, XGBoost, RF**) with residual meta-modeling and weighted ensembles, boosting baseline from **83.2** → **86.4**. • Pioneered **pseudo-labeling** with blend-aware feature weighting, achieving the team-best score of **86.64**. • Designed robust pipelines: target-wise modeling, **Optuna HPO**, clipping/fallback logic, and **soft blending**; integrated physics-informed constraints and **PINN latent features** for exploratory innovation.
TECHNICAL PROJECTS	
Speaker Recognition using iVectors *(CS6300)* *Prof: Hema A Murthy*	• Developed a **Universal Background Model (UBM)** using MFCCs; trained per-speaker **GMMs** initialized from the UBM to capture speaker-dependent characteristics. • Represented each speaker with an **ivector** via Total Variability Space; applied **LDA** and **WCCN** on training data. • Attained **92%** accuracy evaluating on test data of **100 speakers**.
codemap — Codebase Comprehension Tool *(Open Source, Solo)* *github.com/hementhu/codemap*	• Built and open-sourced an interactive **visual map for Python codebases** — understand structure in 20 minutes instead of 2 days; aimed at developers using AI coding tools who generate code they never fully read. • Pure **AST-based static analysis (no LLM — zero hallucination)**: architecture map of files and imports, per-function call graphs with return types, and inferred purpose/reading order.

	• Every claim the tool makes ships with its **evidence — exact numbered source lines**, with docstrings explicitly labelled "documented intent (verify against code)".
Reinforcement Learning Agent – 2048 Game Bot	• Designed and trained RL agents (**DQN, Policy Gradient, PPO**) with custom reward functions (tile merges, monotonicity, board smoothness) for long-term strategy. • Implemented experience replay, target networks, and ε-greedy exploration **to stabilize training**; hyperparameter tuning achieved consistent **>90% win rate** for the 1024 tile with progress toward 2048. • Built the full training pipeline in PyTorch **with logging, evaluation, and reproducible benchmark runs**.
Sentiment Analysis	• Applied the **DistilRoBERTa** transformer for sentiment analysis on user reviews inside a gaming platform; integrated **NLTK** preprocessing. • Attained an **F1-score of 0.91** on training and test sets.
Software Team Head, Krishaka, IITM *(Apr '21 – '23)*	• Led the software team developing a navigation stack by sensor-fusing RGB-D and LiDAR data; implemented **YOLO, ResNet and U-Net** for crop-row segmentation. • Reduced robot localization error by **40%** using point cloud data; achieved IoU of **0.8**.
RGB2Thermal *(EE5791)* *Prof: Kaushik Mitra*	• Adapted a **pix2pix GAN** on 2k RGB images to translate them into accurate thermal images, using a **UNET** to extract semantic attributes. • Tuned the model to a PSNR of **27.29 dB**; stood **2nd out of 30** teams in class.

SOCIAL IMPACT AND EXTRACURRICULAR ACTIVITIES

Zenith Faculties *(Feb '22 – Apr '22)*	• Taught mathematics to school students across grade levels, adapting explanations to each student's pace and foundation. • Counseled students one-on-one to reduce academic stress and improve engagement.
Sports	Selected for **NSO (National Sports Organization) – Fitness**; active participant in chess events.

LEADERSHIP INITIATIVES AND POSITIONS OF RESPONSIBILITY

iBot Club Head CFI, IIT Madras *(Apr '22 – Apr '23)*	• Led a team of **70+ members** across 5 projects solving real-world problems; represented IIT Madras in nationwide robotics competitions and tech fests. • Conceptualized and executed a state-wide **summer school** (120+ registrations) and an institute-level **robotics competition** with 100+ participants.
Events Coordinator, Shaastra *(Dec '21 – Jan '22)*	• Led a team of 4 to conduct a **nationwide CAD Design challenge** sponsored by L&T; part of the jury deciding the winner.
Miscellaneous	**Five-star rated coder** on HackerRank in **Python**.